WEIGHTED FRACTIONAL LANE–EMDEN SYSTEMS: ARBITRARY REAL WEIGHTS AND ALL POSITIVE POWERS

Abstract. We present an extension of the localized kernel virial and interval pressure method to weighted fractional Lane–Emden systems for general solution pairs. The statement allows all real weights and all positive powers, in every integer dimension $N > 2s$, $0 < s < 1$. Continuity at the origin and the usual fractional tail class are explicit. The potential reduction forces admissible origin weights. The product range $pq \leq 1$ is excluded directly. For $pq > 1$ the weighted critical hyperbola enters through universal local energy and dilation. Diagonal cancellation covers $2s \leq 1$; the one-dimensional pressure argument uses a single line.

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1. Introduction and main theorem

Caffarelli and Silvestre [2] realized the fractional Laplacian as a weighted Dirichlet-to-Neumann map for a degenerate elliptic extension problem. Using the extension formulation and moving planes, Quaas and Xia [8] proved a Liouville theorem for the unweighted fractional Lane–Emden system in a restricted range of subcritical exponents. Chen, Li and Li [3] developed a direct moving-plane method for the fractional Laplacian, including maximum principles that yield symmetry and nonexistence results for semilinear equations.

For equations and systems with power weights, Dou and Zhou [4] established Liouville theorems for fractional Hénon problems by relating them to integral equations and applying moving planes in integral form. Li, Yu and Zhang [5] treated fractional Hardy–Hénon systems with singular real weights and possibly different fractional orders in dimensions $N \geq 2$. Their analysis includes distributional extension across the origin, integral representations, and nonexistence results in specified exponent ranges obtained through integral estimates and scaling spheres.

The method in this paper follows the localized virial and source-layer interval construction in [7], with the fractional cancellation developed in [6]. The latter preprint reports the unweighted strict-subcritical result for $N \geq 2$, $0 < s < 1$, and all $p, q > 0$. The present argument treats the weighted equations and also gives a separate proof when $N = 1 > 2s$. Each estimate needed for the weighted system, sublinear powers, and dimension one is proved here. The potential reduction uses the analytic ball Green and Poisson formulas in [1].

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For arbitrary negative weights, the equations are imposed away from the origin, while the solutions are required to extend continuously across the origin. This is part of the solution class. In particular, punctured-space solutions with a singularity at the origin are outside the theorem.

Theorem 1.1 (Weighted fractional subcritical nonexistence)**.** *Let $N \geq 1$ be an integer, $0 < s < 1$, and $N > 2s$. Let $p, q > 0$ and $A, B \in \mathbb{R}$. Assume*

$$\frac{N+A}{p+1} + \frac{N+B}{q+1} > N - 2s.$$

Put $L_s = \{w \in L^1_{\mathrm{loc}}(\mathbb{R}^N) : \int_{\mathbb{R}^N} |w(x)|/(1+|x|^{N+2s})\,dx < \infty\}$. Let

$$u, v \in C(\mathbb{R}^N) \cap C^{1,1}_{\mathrm{loc}}(\mathbb{R}^N \setminus \{0\}) \cap L_s$$

be nonnegative. Suppose, pointwise at every $x \neq 0$,

$$(-\Delta)^s u(x) = |x|^A v(x)^p, \qquad (-\Delta)^s v(x) = |x|^B u(x)^q,$$

where $(-\Delta)^s$ is the standard positive normalized fractional Laplacian, with Fourier symbol $|\xi|^{2s}$, equivalently its normalized principal-value singular integral. Then $u \equiv v \equiv 0$.

Taking $A = B = 0$ gives the unweighted fractional Lane–Emden conclusion for every $p, q > 0$ with $1/(p+1) + 1/(q+1) > (N-2s)/N$ in the same class. The reduction below also shows that $A \leq -2s$, $B \leq -2s$, or $pq \leq 1$ individually excludes a nonzero pair without any subcritical assumption.

The proof proceeds from origin removal and exact potentials to universal source bounds, a symmetrized weighted virial, the interval inequality, and pressure. The one-dimensional and higher-dimensional pressure arguments are separate. The final dilation argument proves the integral theorem and completes the main theorem.

2. Reduction to positive potentials and the small product range

Definition 2.1 (Weighted fractional solution class)**.** Let N be a positive integer, let $0 < s < 1$ with $N > 2s$, and put $\sigma = 2s$ and $\lambda = N - \sigma > 0$. Let $p, q > 0$ and $A, B \in \mathbb{R}$. Define

$$\mathcal{L}_s(\mathbb{R}^N) = \left\{ w \in L^1_{\mathrm{loc}}(\mathbb{R}^N) : \int_{\mathbb{R}^N} \frac{|w(x)|}{1+|x|^{N+2s}}\,dx < \infty \right\}.$$

The normalized whole-space fractional Laplacian is

$$(-\Delta)^s w(x) = C_{N,s}\,\mathrm{PV} \int_{\mathbb{R}^N} \frac{w(x) - w(y)}{|x-y|^{N+2s}}\,dy, \qquad C_{N,s} = \frac{4^s \Gamma(N/2+s)}{\pi^{N/2}|\Gamma(-s)|}.$$

On Schwartz functions we use the Fourier convention $\widehat{w}(\xi) = \int_{\mathbb{R}^N} e^{-ix\cdot\xi} w(x)\,dx$; this operator has symbol $|\xi|^{2s}$. Its Riesz fundamental kernel is denoted by

$$K(x) = c_{N,s}|x|^{-\lambda}, \qquad c_{N,s} = \frac{\Gamma(N/2-s)}{4^s \pi^{N/2}\Gamma(s)}.$$

A solution pair consists of nonnegative functions

$$u, v \in C(\mathbb{R}^N) \cap C^{1,1}_{\mathrm{loc}}(\mathbb{R}^N \setminus \{0\}) \cap \mathcal{L}_s(\mathbb{R}^N)$$

satisfying, pointwise for $x \neq 0$,

$$(-\Delta)^s u(x) = |x|^A v(x)^p, \qquad (-\Delta)^s v(x) = |x|^B u(x)^q.$$

Write $g(x) = |x|^A v(x)^p$ and $f(x) = |x|^B u(x)^q$ for $x \neq 0$, and assign the irrelevant values $g(0) = f(0) = 0$.

Lemma 2.2 (Removal of the origin in the distributional equations)**.** *For every solution pair of 2.1, the sources f, g belong to $L^1_{\mathrm{loc}}(\mathbb{R}^N)$ and the two equations hold on all of $\mathbb{R}^N$ in the distributional sense:*

$$\int_{\mathbb{R}^N} u(-\Delta)^s \varphi = \int_{\mathbb{R}^N} g\varphi, \qquad \int_{\mathbb{R}^N} v(-\Delta)^s \varphi = \int_{\mathbb{R}^N} f\varphi \quad (\varphi \in C_c^\infty(\mathbb{R}^N)).$$

In particular, there is no distribution supported at the origin in either equation.

Proof of 2.2. Use the notation and hypotheses of 2.1. First, the pointwise equations imply the corresponding distributional identities for tests supported away from zero. Indeed, truncate the principal-value integral at $|x-y|=\delta$, use symmetry and Fubini, and then let $\delta \downarrow 0$. On a compact set away from zero the $C^{1,1}$ assumption bounds the symmetrized second difference by $C|x-y|^2$, whose omitted integral is $O(\delta^{2-\sigma})$. The tails are integrable by the $\mathcal{L}_s$ condition. For a smooth compact test φ, the truncated operators satisfy, uniformly for $0<\delta<1$,

$$|(-\Delta)^s_\delta \varphi(x)| \le \frac{C_\varphi}{1+|x|^{N+\sigma}}.$$

On bounded sets this follows from the symmetric second-difference estimate; outside a fixed ball containing the support it follows directly from the kernel integral. Thus dominated convergence against $u \in \mathcal{L}_s$ justifies the limit in $\int \varphi(-\Delta)^s_\delta u = \int u(-\Delta)^s_\delta \varphi$. The same reasoning applies to v.

Choose $\chi \in C_c^\infty(B_2)$ with $0 \le \chi \le 1$ and $\chi = 1$ on B_1, and let $\chi_\varepsilon(x) = \chi(x/\varepsilon)$. For a fixed $\varphi \in C_c^\infty(\mathbb{R}^N)$ set $\psi_\varepsilon = \varphi\chi_\varepsilon$. For $0<\varepsilon<1/4$,

$$|(-\Delta)^s \psi_\varepsilon(x)| \le \begin{cases} C_\varphi \varepsilon^{-\sigma}, & |x| \le 4\varepsilon, \\ C_\varphi \varepsilon^N |x|^{-N-\sigma}, & |x| > 4\varepsilon. \end{cases}$$

To prove the first estimate, rescale by $x=\varepsilon z$. The functions $z \mapsto \varphi(\varepsilon z)\chi(z)$ have uniformly bounded C^2 norms and fixed compact support. The symmetric second difference handles $|z-y| \le 1$, and the bounded function handles its complement. For the second estimate, $\psi_\varepsilon(x)=0$ and its support lies in $B_{2\varepsilon}$, so the defining integral has absolute value at most $C\varepsilon^N(|x|-2\varepsilon)^{-N-\sigma}$.

Continuity makes u bounded on B_1. Splitting the integral into $B_{4\varepsilon}$, $B_1 \setminus B_{4\varepsilon}$, and $\mathbb{R}^N \setminus B_1$ therefore gives

$$\begin{aligned} \int_{\mathbb{R}^N} u|(-\Delta)^s\psi_\varepsilon| &\le C_{u,\varphi}\left[\varepsilon^{N-\sigma} + \varepsilon^N \int_{4\varepsilon}^1 r^{-1-\sigma}\,dr + \varepsilon^N \int_{|x|>1} \frac{u(x)}{|x|^{N+\sigma}}\,dx\right] \\ &\le C_{u,\varphi}\varepsilon^{N-\sigma} \longrightarrow 0. \end{aligned}$$

The same estimate holds with v. The exponent $N-\sigma$ is positive even in dimension one, since that case requires $s<1/2$.

Choose first a nonnegative φ that equals one near zero. Its punctured test $\varphi(1-\chi_\varepsilon)$ yields

$$\int g\varphi(1-\chi_\varepsilon) = \int u(-\Delta)^s\varphi - \int u(-\Delta)^s\psi_\varepsilon.$$

The right-hand side has a finite limit; here $(-\Delta)^s\varphi$ is bounded locally and is $O(|x|^{-N-\sigma})$ at infinity. The left-hand side is nonnegative. Fatou's lemma implies $\int g\varphi < \infty$, and hence local integrability at zero. Local integrability elsewhere follows from continuity of the source. Apply the same reasoning to f. For an arbitrary, possibly signed test, local integrability now permits dominated convergence on the source side, and the displayed cutoff estimate permits the limit on the solution side. This proves the two global identities and excludes every origin defect. □

Lemma 2.3 (Exact finite Riesz representation and admissible weights)**.** *Every solution pair of 2.1 has one of the following two alternatives:*

1. $u \equiv v \equiv 0$*;*
2. $u(x)>0$ *and* $v(x)>0$ *for every* $x \in \mathbb{R}^N$*,* $A, B > -\sigma$*, and the finite continuous potentials satisfy*

$$u(x) = \int_{\mathbb{R}^N} K(x-y)g(y)\,dy, \qquad v(x) = \int_{\mathbb{R}^N} K(x-y)f(y)\,dy \quad (x \in \mathbb{R}^N).$$

In particular,

$$\int_{\mathbb{R}^N} |y|^{-\lambda} g(y)\,dy < \infty, \qquad \int_{\mathbb{R}^N} |y|^{-\lambda} f(y)\,dy < \infty.$$

Proof of 2.3. Use 2.1 and the global distributional equations proved in 2.2. We use Bucur's fundamental-solution and compact-source potential results [1, Theorems 2.3 and 2.8], the ball Poisson and Green representations [1, Theorems 2.10 and 3.2], and the explicit kernels and normalization constants [1, Theorems 3.1, 3.3 and 3.5]. These locators refer to arXiv version 4 and apply for $0 < s < 1$ in every positive integer dimension; here $N > 2s$. The Green sources below are smooth and compactly supported. The Poisson theorem allows continuous exterior data in $\mathcal{L}_s$, as used below. The constants agree with our normalization; Bucur's $e^{-2\pi i x\cdot\xi}$ Fourier convention changes the symbol, rather than the physical operator.

We shall also use the ball Poisson kernel

$$P_R(x,z) = \frac{\Gamma(N/2)\sin(\pi s)}{\pi^{N/2+1}} \frac{(R^2-|x|^2)^s}{(|z|^2-R^2)^s|x-z|^N}, \qquad |x| < R < |z|.$$

For $x, y \in B_R$, $x \neq y$, the zero-exterior Green kernel is

$$G_R(x,y) = \frac{\Gamma(N/2)}{4^s\pi^{N/2}\Gamma(s)^2}|x-y|^{-\lambda}\int_0^{\rho_R(x,y)} \frac{t^{s-1}}{(1+t)^{N/2}}\,dt, \qquad \rho_R(x,y) = \frac{(R^2-|x|^2)(R^2-|y|^2)}{R^2|x-y|^2}.$$

The kernel is nonnegative. Fix a smooth compactly supported function h in $B_R \setminus \{0\}$ with $0 \le h \le g$, and put $V_R(x) = \int_{B_R} G_R(x,y)h(y)\,dy$ inside the ball and zero outside. The Green theorem gives global continuity and the pointwise equation $(-\Delta)^s V_R = h$ in B_R. Its interior smoothness can be seen directly: extend h by zero and put $H = K * h$. Differentiation can be transferred to the smooth compact source, so $H \in C^\infty(\mathbb{R}^N)$; the compact-source potential theorem also gives $H \in \mathcal{L}_s$. The defining Green construction and Tonelli yield

$$V_R(x) = H(x) - \int_{|z|>R} P_R(x,z)H(z)\,dz, \qquad x \in B_R.$$

Since $H \in C(\mathbb{R}^N) \cap \mathcal{L}_s$, the interior smoothness of its Poisson extension established in [1, proof of Theorem 2.10] applies. The displayed identity therefore gives $V_R \in C^\infty(B_R)$. Its global continuity and compact support also give $V_R \in \mathcal{L}_s$. Thus $w = u - V_R$ is continuous, lies in $\mathcal{L}_s$, is nonnegative outside B_R, and has $(-\Delta)^s w \ge 0$ in $B_R \setminus \{0\}$.

For $\varepsilon > 0$, add the barrier $\varepsilon K(x)$. This is positive, lies in $\mathcal{L}_s$, is s-harmonic off zero, and diverges to $+\infty$ at zero. If $w + \varepsilon K$ had a negative value, it would attain a negative minimum in $B_R \setminus \{0\}$: continuity excludes the boundary, and the barrier excludes zero. At the minimum its fractional Laplacian is strictly negative. Indeed every difference from the minimum is nonpositive, and the exterior of B_R has positive measure and nonnegative values. The principal value is legitimate at this interior point by local $C^{1,1}$ regularity. This contradicts its nonnegative fractional Laplacian. Thus $w + \varepsilon K \ge 0$ off zero for every $\varepsilon > 0$. For each fixed $x \neq 0$, let $\varepsilon \downarrow 0$ to obtain $u(x) \ge V_R(x)$; continuity then gives the inequality also at $x = 0$.

A nonnegative continuous function on the punctured ball admits smooth compact minorants h_j with $0 \le h_j \le g$ and $h_j \to g$ pointwise off zero. Here is an explicit construction. For all sufficiently large j, choose a smooth compact cutoff ζ_j on the punctured ball, with $0 \le \zeta_j \le 1$, that equals one where $1/j \le |x| \le R - 1/j$. On a compact neighborhood of its support, approximate g uniformly by a smooth function a_j with error at most δ_j, where $0 < \delta_j \to 0$. Such approximations follow by mollifying a continuous compact extension from that neighborhood. Choose a smooth function $T_\delta : \mathbb{R} \to [0,\infty)$ such that

$$0 \le T_\delta(t) \le \max(t,0), \qquad |T_\delta(t) - \max(t,0)| \le \delta,$$

and $T_\delta(t) = 0$ for $t \le 0$. For example, integrate a smooth nondecreasing cutoff from zero to one over the interval $[0,\delta]$. Then $h_j = \zeta_j T_{\delta_j}(a_j - \delta_j)$ is smooth, compactly supported away from zero, and satisfies $0 \le h_j \le g$. At each fixed punctured point its error tends to zero. No monotonicity of this sequence is needed. Fatou's lemma gives

$$\int_{B_R} G_R(x,y)g(y)\,dy \le u(x) \quad (x \in B_R).$$

For fixed x, y, $\rho_R(x,y)$ increases to infinity as R increases beyond $\max(|x|,|y|)$. Indeed its numerator divided by R^2 is $R^2 - |x|^2 - |y|^2 + |x|^2|y|^2/R^2$, whose derivative is positive in this

range. The beta integral is

$$\int_0^\infty \frac{t^{s-1}}{(1+t)^{N/2}}\,dt = \frac{\Gamma(s)\Gamma(N/2-s)}{\Gamma(N/2)}.$$

Thus $G_R(x,y) \uparrow K(x-y)$, where kernels are set to zero when either point is outside the ball. Monotone convergence proves the pointwise finite inequalities

$$U(x) := \int K(x-y)g(y)\,dy \le u(x), \qquad V(x) := \int K(x-y)f(y)\,dy \le v(x) \quad (x \in \mathbb{R}^N).$$

If either solution vanishes identically, its equation away from zero forces the other to vanish there and continuity forces it to vanish at zero. Otherwise each source is positive on a nonempty open set away from zero. The positive kernel consequently gives $U, V > 0$ everywhere, and therefore $u, v > 0$ everywhere. In particular $u(0), v(0) > 0$. On a sufficiently small ball, $v \ge v(0)/2$. Finiteness of $U(0)$ then gives

$$\infty > U(0) \ge c\int_0^r t^{A+\sigma-1}\,dt,$$

which forces $A + \sigma > 0$. Applying the same argument to $V(0)$ forces $B + \sigma > 0$. The displayed inequalities at zero also establish both source-tail moments in the statement.

We record continuity of U, V, including at zero, before identifying the harmonic remainders. Fix $x_0 \neq 0$ and $0 < r_0 < |x_0|/4$. The source g is bounded on $B_{2r_0}(x_0)$, so translation continuity in L^1 of the kernel restricted to B_{3r_0} gives continuity of this local contribution for $x \in B_{r_0}(x_0)$. On the remaining bounded source region the distance from x is at least r_0, and local integrability of g, including at zero, gives an integrable majorant. For sufficiently large $|y|$, the integrand is bounded uniformly in this ball by $C|y|^{-\lambda}g(y)$, whose integral is finite. Dominated convergence handles these last two regions. This proves continuity of U away from zero, and the same argument applies to V.

Near zero, $g(y) \le C|y|^A$. For $|x| \le r$,

$$\int_{|y|\le 2r} |x-y|^{-\lambda}|y|^A\,dy \le C_A r^{A+\sigma} \longrightarrow 0.$$

If $A \ge 0$, bound $|y|^A$ by $(2r)^A$ and integrate the kernel. If $-\sigma < A < 0$ and $x \neq 0$, split at $|y| = 2|x|$. On the inner region, rescaling $y = |x|z$ leaves a uniformly finite integral: near $z = 0$ one uses $A > -N$, and near $z = x/|x|$ one uses $\lambda < N$, with these two singularities separated. Its prefactor is $|x|^{A+\sigma} \le r^{A+\sigma}$. On the outer region, $|x-y| \ge |y|/2$ and the radial integral is bounded by $Cr^{A+\sigma}$. The case $x = 0$ is the same radial integral directly. This proves the estimate. Outside B_{2r}, dominated convergence handles $x \to 0$ for each fixed r; the estimate handles the discarded region uniformly, so U is continuous at zero. The argument with B proves continuity of V.

Since $0 \le U \le u$ and $0 \le V \le v$, both potentials belong to $\mathcal{L}_s$. For a compact smooth test φ,

$$\int U(x)|(-\Delta)^s\varphi(x)|\,dx < \infty.$$

Tonelli applied to the absolute integrand therefore justifies Fubini in the signed pairing. The fundamental-solution identity yields

$$\int U(-\Delta)^s\varphi = \int g(y)\left[\int K(x-y)(-\Delta)^s\varphi(x)\,dx\right]dy = \int g\varphi.$$

Hence $h = u - U$ is a nonnegative continuous function in $\mathcal{L}_s$ that is globally distributionally s-harmonic.

Here is a direct Liouville argument valid also when $N = 1$. Choose $\theta \in C_c^\infty(B_1)$ with $\theta \ge 0$ and $\int\theta = 1$, set $\theta_\varepsilon(x) = \varepsilon^{-N}\theta(x/\varepsilon)$, and let $h_\varepsilon = h * \theta_\varepsilon$. Put $\widetilde{\theta}_\varepsilon(x) = \theta_\varepsilon(-x)$. The weights $1 + |x|^{N+\sigma}$ and $1 + |x+a|^{N+\sigma}$ are comparable uniformly for bounded a. Consequently h_ε is

smooth, nonnegative, and belongs to $\mathcal{L}_s$. For $\varphi \in C_c^\infty(\mathbb{R}^N)$, absolute Fubini and translation invariance give

$$\begin{aligned}\int h_\varepsilon(-\Delta)^s\varphi &= \int h\,[\widetilde{\theta}_\varepsilon * ((-\Delta)^s\varphi)]\\ &= \int h\,(-\Delta)^s(\widetilde{\theta}_\varepsilon * \varphi) = 0.\end{aligned}$$

The convolved test is again smooth and compactly supported. The absolute pairings follow from the fractional decay of the test and the weighted tail of h. Thus h_ε is distributionally s-harmonic. On compact sets its symmetric second difference is bounded by $C|z|^2$. The tail condition then makes its pointwise fractional Laplacian well defined and continuous. The same truncated self-adjoint calculation as in 2.2 identifies this function with the distributional fractional Laplacian, so it vanishes pointwise.

By uniqueness in the ball Poisson theorem [1, Theorem 2.10], h_ε equals the Poisson extension of its exterior values in every ball. In centered coordinates this reads

$$h_\varepsilon(x_0 + x) = \int_{|y|>R} P_R(x,y)h_\varepsilon(x_0+y)\,dy, \qquad |x| < R.$$

For $|x| \le R/2$ and $|y| > R$, differentiation of that kernel gives

$$P_R(x,y) \le 2^N P_R(0,y), \qquad |\nabla_x P_R(x,y)| \le \frac{C}{R}P_R(0,y).$$

Here C depends only on N, s. The center kernel is integrable against the translated exterior data: continuity controls its sphere singularity of exponent $s < 1$, and $\mathcal{L}_s$ controls infinity. Dominated differentiation is therefore valid. At the center the sharper bound is $|\nabla_x P_R(0,y)| = N|y|^{-1}P_R(0,y) \le (N/R)P_R(0,y)$. Nonnegativity and the representation give

$$|\nabla h_\varepsilon(x_0)| \le \frac{N}{R}h_\varepsilon(x_0).$$

Let $R \to \infty$. The gradient vanishes at every x_0, so h_ε is constant. Letting $\varepsilon \downarrow 0$ and using continuity proves $h \equiv c$ with $c \ge 0$. The same reasoning gives $v - V \equiv d$ with $d \ge 0$.

Finally, if $d > 0$, then $v \ge d$ everywhere and

$$U(0) \ge c_{N,s}d^p \int_{|y|>1} |y|^{A-\lambda}\,dy = c_{N,s}|\mathbb{S}^{N-1}|d^p \int_1^\infty r^{A+\sigma-1}\,dr = \infty,$$

because $A + \sigma > 0$. This contradicts $U(0) < \infty$. Thus $d = 0$. If $c > 0$, the corresponding argument for $V(0)$ and $B + \sigma > 0$ gives a contradiction; hence $c = 0$. The exact representations follow. No subcritical inequality, no restriction $p, q \ge 1$, and no condition on pq was used. □

Lemma 2.4 (Nonexistence when pq is at most one)**.** *Every solution pair of 2.1 with $pq \le 1$ is identically zero, for arbitrary real A, B and every integer $N > 2s$. No subcritical inequality is required in this case.*

Proof of 2.4. Suppose the pair is nonzero. By 2.3, u, v are strictly positive everywhere, their Riesz representations are exact, and $a_0 = \sigma + A > 0$ and $b_0 = \sigma + B > 0$. For $R > 0$ define the positive finite minima

$$U_R = \min_{\overline{B_R}} u, \qquad V_R = \min_{\overline{B_R}} v.$$

For $x \in \overline{B_R}$ and $y \in B_R \setminus B_{R/2}$, $K(x-y) \ge cR^{-\lambda}$ except on the irrelevant diagonal. The exact first potential and $v(y) \ge V_R$ consequently give

$$u(x) \ge cR^{-\lambda}V_R^p \int_{B_R\setminus B_{R/2}} |y|^A\,dy = c_A R^{\sigma+A}V_R^p.$$

The annular integral is a positive finite constant times R^{N+A} for every real A. Taking the minimum and repeating the calculation for the second potential gives

$$U_R \ge c_A R^{a_0}V_R^p, \qquad V_R \ge c_B R^{b_0}U_R^q.$$

All constants are positive and independent of R. Substituting the second inequality into the first, and dividing by the positive U_R^{pq}, yields

$$U_R^{1-pq} \geq cR^{a_0+pb_0}.$$

If $pq = 1$, the left side is one, whereas the right side diverges as $R \to \infty$. If $pq < 1$, then $U_R \leq u(0)$ makes the left side bounded by $u(0)^{1-pq}$, while the right side again diverges. Both cases contradict $a_0 + pb_0 > 0$, proving the claim. □

3. Riesz systems and universal source estimates

Definition 3.1 (Weighted Riesz system and localization notation)**.** Fix an integer $N \geq 1$, real numbers $0 < \sigma < N$, $p, q > 0$ with $pq > 1$, real weights $A, B > -\sigma$, and a constant $c > 0$. Put

$$\lambda = N - \sigma, \qquad K(z) = c|z|^{-\lambda} \quad (z \neq 0), \qquad f(x) = |x|^B u(x)^q, \quad g(x) = |x|^A v(x)^p \quad (x \neq 0).$$

Set $f(0) = g(0) = 0$, which does not change any integral. A solution in this section means finite continuous functions $u, v : \mathbb{R}^N \to (0, \infty)$ satisfying, pointwise,

$$u(x) = \int_{\mathbb{R}^N} K(x-y)g(y)\,dy, \qquad v(x) = \int_{\mathbb{R}^N} K(x-y)f(y)\,dy.$$

Write $B_R = \{x : |x| < R\}$ and set

$$\alpha = \frac{\sigma + A + p(\sigma + B)}{pq - 1}, \quad \beta = \frac{\sigma + B + q(\sigma + A)}{pq - 1}, \quad a = \frac{1}{q+1}, \quad b = \frac{1}{p+1}.$$

The nonnegative interaction measure is

$$d\pi(x, y) = f(x)g(y)K(x - y)\,dx\,dy.$$

For the fixed cutoff choose the positive integers

$$a_0 = N + 3, \qquad k = N(a_0 + 1), \qquad L = k + N + 2.$$

Writing $t_+ = \max(t, 0)$, define

$$d(x) = \min\{1, (2-|x|)_+\}, \quad \eta(x) = \Big[\big(1 - (|x| - 1)_+^3\big)_+\Big]^L, \quad w(x) = -x\cdot\nabla\eta(x), \quad X(x) = x\eta(x).$$

For $x \neq 0$ set $\omega_x = x/|x|$, choosing any unit vector for ω_0. For a unit vector $e \in \mathbb{S}^{N-1}$, put

$$D_e(x, y) = \frac{((x - y)\cdot e)^2}{|x - y|^2}, \quad D(x, y) = D_{\omega_x}(x, y), \quad \Phi(x, y) = \frac{(X(x) - X(y))\cdot(x - y)}{|x - y|^2}.$$

Values on the diagonal do not affect any integral. Define

$$E_f = \int \eta f u, \quad E_g = \int \eta g v, \quad E = E_f + E_g, \qquad W_f = \int w f u, \quad W_g = \int w g v.$$

A constant called universal may depend on N, σ, p, q, A, B, c and displayed auxiliary parameters, but never on the solution. All integrations are Lebesgue integrations.

Lemma 3.2 (Universal source and potential estimates)**.** *In the setting of 3.1 there is a universal constant C such that every solution satisfies*

$$\int_{B_R} f \leq CR^{N-\sigma-\beta}, \qquad \int_{B_R} g \leq CR^{N-\sigma-\alpha} \qquad (R > 0),$$

$$\int_{B_4} (f + g + u + v) \leq C,$$

and

$$\int_{|x-y|\geq\ell} K(x - y)(f(y) + g(y))\,dy \leq C\ell^{-\lambda} \qquad (x \in B_2, 0 < \ell \leq 1).$$

The marginals of π are $f u\,dx$ and $g v\,dy$. The quantities E_f, E_g, W_f, W_g are finite.

Proof of 3.2. Use the notation of 3.1. The source masses $M_f = \int_{B_1} f$ and $M_g = \int_{B_1} g$ are strictly positive and finite: u, v are positive and bounded on $\overline{B}_1$, and $A, B > -\sigma > -N$. For distinct $x, y \in B_1$, $K(x-y) \geq C_0 := c2^{-\lambda} > 0$. The potential equations therefore imply

$$u(x) \geq C_0 M_g, \quad v(x) \geq C_0 M_f \quad (x \in B_1),$$

since the excluded diagonal has measure zero. Raising to the positive powers and integrating against the weights gives

$$M_f \geq C_f M_g^q, \qquad M_g \geq C_g M_f^p,$$
$$C_f = C_0^q \int_{B_1} |x|^B \, dx, \quad C_g = C_0^p \int_{B_1} |x|^A \, dx.$$

Both coefficients are positive universal constants. Substitution and division by the positive source masses yield

$$M_f^{pq-1} \leq (C_f C_g^q)^{-1}, \qquad M_g^{pq-1} \leq (C_g C_f^p)^{-1}.$$

Since $pq > 1$, these are universal upper bounds on both masses. This argument uses only monotonicity of positive powers, and therefore also applies when either p or q is below one.

The explicit exponents in 3.1 are positive, since $\sigma + A, \sigma + B > 0$, and direct algebra gives $\alpha + \sigma + A = p\beta$ and $\beta + \sigma + B = q\alpha$. Homogeneity of K shows that

$$u_R(x) = R^\alpha u(Rx), \qquad v_R(x) = R^\beta v(Rx)$$

is again a solution: changing variables $z = Ry$ in its first potential gives

$$\int K(x-y)|y|^A v_R(y)^p \, dy = R^{p\beta-\sigma-A} \int K(Rx-z)|z|^A v(z)^p \, dz = R^\alpha u(Rx).$$

The second equation follows from $q\alpha - \sigma - B = \beta$. The rescaled functions remain finite, continuous and positive, and all parameters, including c, are unchanged. Their sources obey $f_R(x) = R^{\beta+\sigma} f(Rx)$ and $g_R(x) = R^{\alpha+\sigma} g(Rx)$. Applying the unit-ball estimates with the same universal constants gives

$$R^{\beta+\sigma-N} \int_{B_R} f = \int_{B_1} f_R \leq C, \qquad R^{\alpha+\sigma-N} \int_{B_R} g = \int_{B_1} g_R \leq C.$$

These are the asserted source bounds for every $R > 0$.

For $x \in B_4$ and $j \geq 3$, the distance to the annulus $2^j \leq |y| < 2^{j+1}$ is at least $2^j - 4 \geq 2^{j-1}$. Its contribution to the g potential is therefore at most

$$C2^{-j\lambda} \int_{B_{2^{j+1}}} g \leq C2^{-j\alpha}.$$

The analogous f contribution is at most $C2^{-j\beta}$. Since $\alpha, \beta > 0$, summing the geometric series gives the uniform tail estimate

$$\sup_{x \in B_4} \int_{|y| \geq 8} K(x-y)(f(y) + g(y)) \, dy \leq C.$$

The sources on B_8 have universal total mass, and

$$\sup_{y \in B_8} \int_{B_4} K(x-y) \, dx \leq c \int_{B_{12}} |z|^{-\lambda} \, dz < \infty,$$

since $\lambda < N$. Tonelli's theorem bounds the integral over B_4 of the two near-source potentials by $C \int_{B_8} (f+g) \leq C$; the uniform tail bound handles the remaining contribution. Thus $\int_{B_4} (u+v) \leq C$. Together with the source bounds this proves the second assertion. For $x \in B_2$ and $0 < \ell \leq 1$, the same splitting gives

$$\int_{|x-y| \geq \ell} K(x-y)(f(y) + g(y)) \, dy \leq c\ell^{-\lambda} \int_{B_8} (f+g) + C \leq C\ell^{-\lambda},$$

because $\ell^{-\lambda} \geq 1$. This proves the third assertion.

For any nonnegative Borel function ψ, Tonelli and the first potential equation give

$$\iint \psi(x)\,d\pi(x,y) = \int \psi(x) f(x) \left[\int K(x-y) g(y)\,dy \right] dx = \int \psi f u.$$

The second marginal is $gv\,dy$ by the second equation and $K(x-y) = K(y-x)$. These identities hold also when an integral is infinite. The marginal densities $fu = |x|^B u^{q+1}$ and $gv = |x|^A v^{p+1}$ are locally integrable, since $A, B > -N$ and u, v are locally bounded. The cutoffs η, w are bounded and supported in $\overline{B}_2$, so all four localized energies are finite. □

4. Regularity, the weighted virial, and localization

Lemma 4.1 (Regularity and principal-value gradients away from the weight origin)**.** *In the setting of 3.1, the functions u, v, f, g are C^1 on $\mathbb{R}^N \setminus \{0\}$. For each $x \neq 0$,*

$$\nabla u(x) = \lim_{\varepsilon \downarrow 0} \int_{|x-y|>\varepsilon} \nabla K(x-y) g(y)\,dy, \qquad \nabla v(x) = \lim_{\varepsilon \downarrow 0} \int_{|x-y|>\varepsilon} \nabla K(x-y) f(y)\,dy.$$

Both limits are locally uniform away from zero. If $\sigma > 1$, the gradient integrals converge absolutely there. For every $\sigma > 0$, the singular part becomes absolutely integrable after replacing $g(y)$ by $g(y) - g(x)$, or $f(y)$ by $f(y) - f(x)$, inside a fixed ball centered at x. The far-source integrals remain unsubtracted.

Proof of 4.1. Use the positive finite continuous potential system of 3.1. Fix a ball whose closure avoids zero and a smaller concentric ball. A smooth cutoff of either source, equal to one on a neighborhood of the smaller ball and supported in the larger ball, separates the potential into a compact near part and a far part. The far part is smooth on a still smaller ball: every derivative of $K(x-y)$ there is bounded by a fixed multiple of $K(x_0 - y)$ on the support of the far source, where x_0 is the center. This follows from the positive separation and the homogeneity of the kernel. Its integral is finite by the exact potential equation at x_0. Dominated differentiation proves the assertion for every derivative of the far part.

Here is an elementary smoothing argument for the near part. Let ζ be a compactly supported smooth radial cutoff equal to one near zero and put $\kappa(z) = K(z)\zeta(z)$. The kernel κ is integrable, because $\lambda = N - \sigma < N$. It suffices to examine $\kappa * G$, where G is a compactly supported local cutoff of the source. The difference between this expression and the near potential has a smooth kernel near the points under consideration. If $\sigma > 1$, then $\nabla\kappa \in L^1$, and the derivative of $\kappa * G$ is continuous by translation continuity in L^1 against the bounded function G. Thus the potential is C^1 in this case.

Suppose now that $0 < \sigma \leq 1$. Translation invariance of the integral of κ gives

$$\int_{\mathbb{R}^N} [\kappa(z+h) - \kappa(z)]\,dz = 0.$$

If G is globally Hölder continuous with exponent $0 < \theta < 1$, then

$$(\kappa * G)(x+h) - (\kappa * G)(x) = \int [\kappa(z+h) - \kappa(z)]\,[G(x-z) - G(x)]\,dz.$$

For $0 < |h| < 1/4$, split this integral into $|z| \leq 2|h|$ and its complement. On the first region, $|G(x-z) - G(x)| \leq C|h|^\theta$, and the integrals of $|\kappa(z+h)| + |\kappa(z)|$ are at most $C|h|^\sigma$. On the second region, the mean value theorem bounds the kernel difference by $C|h||z|^{\sigma-N-1}$ near zero; the contribution from the remaining fixed annulus is at most $C|h|$. Consequently

$$|(\kappa * G)(x+h) - (\kappa * G)(x)| \leq C|h|^{\sigma+\theta} + C|h| \int_{2|h|}^{R_0} r^{\sigma+\theta-2}\,dr + C|h|,$$

where R_0 is fixed. If $\sigma + \theta < 1$, this is at most $C|h|^{\sigma+\theta}$; at equality it is at most $C|h|(1 + |\log|h||)$, hence at most $C_t|h|^t$ for every $t < 1$. The same splitting with $|G(x-z) - G(x)| \leq 2\|G\|_\infty$ proves that a bounded source gives a potential of every Hölder order strictly less than $\min(\sigma, 1)$.

Apply these estimates on nested balls avoiding zero. On each such compact set, positivity and continuity bound u and v above and away from zero. Therefore $t \mapsto t^p$ and $t \mapsto t^q$ have

bounded derivatives on their respective local ranges, including when a power is less than one. Multiplication by the smooth weights $|x|^A$ and $|x|^B$ preserves the local Hölder exponent. Smooth cutoffs of these sources extend to globally Hölder functions. To make the number of iterations explicit, put

$$J = \min\{j \in \mathbb{N}_{>0} : j\sigma/2 > 1 - \sigma\}, \qquad \theta_j = j\sigma/2 \quad (1 \le j \le J).$$

Choose $J + 1$ nested balls between any prescribed smaller ball and a larger ball whose closure avoids zero. The bounded-source estimate first gives u, v, and hence f, g, exponent $\theta_1 = \sigma/2$ on a smaller ball. If $j < J$, then $\theta_j \le 1 - \sigma$ and $\theta_{j+1} < \theta_j + \sigma$. When $\theta_j + \sigma = 1$, the logarithmic estimate supplies every exponent below one; otherwise the strict-gain estimate applies. In both cases it supplies θ_{j+1} on the next smaller ball. Since

$$1 - \sigma < \theta_J \le 1 - \sigma/2 < 1,$$

this finite iteration gives the local sources a Hölder exponent $\theta > 1 - \sigma$.

For such a source, we prove that $\kappa * G$ is C^1, with gradient

$$\int_{\mathbb{R}^N} \nabla\kappa(z)\,[G(x - z) - G(x)]\,dz.$$

The local integrand is bounded by $C|z|^{\sigma-N-1+\theta}$, whose radial integral $C\int_0^1 r^{\sigma+\theta-2}\,dr$ is finite. The remainder is integrable because κ is compactly supported, and dominated convergence proves continuity of the displayed function.

Mollify G to smooth compactly supported functions G_j with uniformly bounded Hölder seminorms and $G_j \to G$ uniformly. For fixed j, differentiate $\kappa * G_j$ by placing the derivative on G_j, which is justified by $\kappa \in L^1$. Integrate by parts outside B_ε. Radial symmetry gives $\int_{|z|>\varepsilon} \nabla\kappa(z)\,dz = 0$ and cancels the constant $G_j(x)$ in the inner boundary term. The remaining boundary term and the omitted integral of $\kappa(z)\nabla G_j(x - z)$ over B_ε are $O_j(\varepsilon^\sigma)$. Letting $\varepsilon \downarrow 0$ therefore gives the displayed gradient formula for G_j. The potentials converge uniformly because $\kappa \in L^1$. For fixed small $r > 0$, the sup norm of the difference between their gradients and the displayed function for G is at most

$$Cr^{\sigma+\theta-1} + 2\|G_j - G\|_\infty \int_{|z|\ge r} |\nabla\kappa(z)|\,dz.$$

First let $j \to \infty$, then $r \downarrow 0$. Thus the gradients converge uniformly as well. Passing to the limit in the fundamental theorem of calculus along coordinate segments shows that $\kappa * G$ is C^1 with the asserted gradient. This proves $u, v \in C^1$ away from zero in the remaining cases. The chain rule now gives $f, g \in C^1$ there.

Finally, let $F \subset \mathbb{R}^N \setminus \{0\}$ be compact and choose $r_0 > 0$ so that its closed r_0-neighborhood avoids zero. The sources and their first derivatives are bounded on that neighborhood. On a centered ball, the odd kernel gradient has zero integral over every concentric annulus. Hence for $0 < \varepsilon < r_0$,

$$\int_{\varepsilon<|x-y|<r_0} \nabla K(x - y)g(y)\,dy = \int_{\varepsilon<|x-y|<r_0} \nabla K(x - y)[g(y) - g(x)]\,dy.$$

The integral over $|x - y| < \varepsilon$ after this subtraction is bounded uniformly for $x \in F$ by

$$C_F \int_0^\varepsilon r^{\sigma-1}\,dr = \frac{C_F}{\sigma}\varepsilon^\sigma.$$

The far integral satisfies

$$\int_{|x-y|\ge r_0} |\nabla K(x - y)|g(y)\,dy \le \frac{\lambda}{r_0}u(x),$$

which is uniformly bounded on F. When the derivatives of the near and far potentials are recombined, the terms containing derivatives of the smooth kernel cutoff cancel. The resulting derivative is precisely the stated principal-value integral. The $C_F\varepsilon^\sigma/\sigma$ bound on its omitted local part proves locally uniform convergence. Replacing g, u by f, v proves the other formula.

If $\sigma > 1$, local integrability of $|\nabla K|$ makes the unsubtracted near integral absolutely convergent as well. □

Lemma 4.2 (Weighted virial identity and localization for every positive Riesz order)**.** *In the setting of 3.1, the following identity has an absolutely convergent right-hand side:*

$$a(N+B)E_f + b(N+A)E_g - aW_f - bW_g = \lambda \iint \Phi(x,y)\, d\pi(x,y).$$

For each $\varepsilon > 0$ there is a universal constant C_ε such that

$$|E_f - E_g| + \iint |\Phi(x,y) - \eta(x) + w(x)D(x,y)|\, d\pi(x,y) \le \varepsilon E + C_\varepsilon.$$

Proof of 4.2. Use the notation of 3.1, the finite energies and marginal identities of 3.2, and the off-origin regularity and principal-value gradient formulas of 4.1. We first prove a symmetrized identity for a C^1 compactly supported vector field Y whose support avoids zero. For a diagonal truncation parameter $h > 0$, each separate double integral below is absolutely convergent, since $|\nabla K(x-y)| \le \lambda h^{-1} K(x-y)$ on $|x-y| > h$ and the compact marginal masses are finite. Fubini and interchange of variables give

$$\begin{aligned} &-\int f(x)Y(x) \cdot \int_{|x-y|>h} \nabla K(x-y) g(y)\, dy\, dx \\ &\quad -\int g(y)Y(y) \cdot \int_{|x-y|>h} \nabla K(y-x) f(x)\, dx\, dy \\ &= \lambda \iint_{|x-y|>h} \frac{(Y(x)-Y(y))\cdot(x-y)}{|x-y|^2}\, d\pi(x,y). \end{aligned}$$

The two left-hand integrals converge to the corresponding gradient integrals as $h \downarrow 0$, by the locally uniform convergence in 4.1 and the integrability of the sources on the support of Y. On the right,

$$\left| \frac{(Y(x)-Y(y))\cdot(x-y)}{|x-y|^2} \right| \le \|\nabla Y\|_\infty \big(\mathbf{1}_{\operatorname{supp} Y}(x) + \mathbf{1}_{\operatorname{supp} Y}(y) \big).$$

This follows from the Lipschitz bound and the vanishing of the numerator when both endpoints lie outside the support. The right-hand side is integrable by the compact marginals from 3.2. Dominated convergence therefore gives

$$-\int fY \cdot \nabla u - \int gY \cdot \nabla v = \lambda \iint \frac{(Y(x)-Y(y))\cdot(x-y)}{|x-y|^2}\, d\pi.$$

This argument combines the two kernel gradients before taking the diagonal limit and does not claim their separate absolute convergence when $\sigma \le 1$.

To remove the weight origin, fix a smooth radial function χ that is zero on $[0,1]$, one on $[2,\infty)$, and has values in $[0,1]$. Put $\chi_\rho(x) = \chi(|x|/\rho)$ and $X_\rho = \chi_\rho X$, where $0 < \rho < 1/4$. This vector field is supported in $\overline{B}_2$ and vanishes on B_ρ. Since $X = x\eta$ and $|x| \asymp \rho$ on the support of $\nabla\chi_\rho$, the quantities $\|\nabla X_\rho\|_\infty$ are bounded independently of ρ. Weighted integration by parts, entirely away from zero, yields

$$-\int fX_\rho \cdot \nabla u = a \int \chi_\rho[(N+B)\eta - w] fu + a \int (X \cdot \nabla\chi_\rho) fu.$$

Here we used the chain rule for u^{q+1} and $\operatorname{div}(|x|^B X) = |x|^B[(N+B)\eta - w]$. The coefficient $X \cdot \nabla\chi_\rho$ is uniformly bounded and supported in $B_{2\rho}$, so the last term tends to zero by the local integrability of the marginal fu established in 3.2. Dominated convergence applies to the first term. The same argument with g, v, A, b gives the other weighted integration by parts limit. Thus their sum tends to

$$a(N+B)E_f + b(N+A)E_g - aW_f - bW_g.$$

On the symmetrized right-hand side the kernel for X_ρ is uniformly bounded by the preceding Lipschitz estimate. Its difference from Φ vanishes unless $x \in B_{2\rho}$ or $y \in B_{2\rho}$, and is uniformly bounded. Its integral is therefore at most

$$C\left(\int_{B_{2\rho}} fu + \int_{B_{2\rho}} gv\right) \longrightarrow 0.$$

This proves the weighted identity. The same Lipschitz and compact-marginal domination proves absolute convergence of the integral containing Φ. The argument requires no separate kernel-gradient integral at the origin.

We next prove localization. The cutoff in 3.1 is C^2, equals one on B_1, and is supported in $\overline{B}_2$. On $1 < |x| < 2$, putting $t = |x| - 1$ gives

$$1 - t^3 = d(x)(1 + t + t^2).$$

Differentiating its Lth power therefore proves

$$d^L \le \eta \le Cd^L, \qquad |\nabla\eta| + w \le Cd^{L-1}, \qquad |\nabla^2\eta| + |\nabla w| + |\nabla^2 X| \le Cd^{L-2}, \qquad dw \le C\eta.$$

These statements extend across both boundary spheres: the cubic at radius one makes its first two derivatives agree with those of the constant inner cutoff, and the power L makes them vanish at radius two. The apparent radial-coordinate singularity at zero is absent because the cutoff is constant there. The same reasoning is valid in dimension one.

Fix $0 < \delta < 1/4$ and put $\ell(z) = \delta d(z)^{a_0}$ for $d(z) > 0$. A pair is called close if $|x - y| \le \ell(z)$ for at least one endpoint $z \in \{x, y\}$ with $d(z) > 0$, and far otherwise. Since d is 1-Lipschitz and $\ell(z) \le \delta d(z)$, on a close pair every point of the segment between x and y has its d-value between $(1 - \delta)d(z)$ and $(1 + \delta)d(z)$. In particular the segment is in B_2. Its endpoints satisfy the same comparability.

The derivative matrix of X is

$$\nabla X(x) = \eta(x)\,\mathrm{Id} - w(x)\omega_x \otimes \omega_x,$$

where the second term is zero on B_1. For $x \neq y$, write $e_{xy} = (x - y)/|x - y|$. The fundamental theorem of calculus along the segment gives

$$\Phi(x, y) - \eta(x) + w(x)D(x, y) = \int_0^1 e_{xy} \cdot \big[\nabla X(x + t(y - x)) - \nabla X(x)\big] e_{xy}\, dt.$$

On a close pair the segment has d-values comparable to $d(z)$, so the integrand is bounded by $C|x - y|d(z)^{L-2}$. The analogous first-derivative estimate for η gives

$$|\Phi - \eta(x) + w(x)D| \le C\delta d(z)^{L-2+a_0}, \qquad |\eta(x) - \eta(y)| \le C\delta d(z)^{L-1+a_0}.$$

Because $a_0 \ge 2$ and $0 < d(z) \le 1$, both bounds are at most $C\delta d(z)^L$. Also

$$d(z)^L \le (1 - \delta)^{-L} d(x)^L \le (1 - \delta)^{-L}\eta(x),$$

and the same estimate holds with y in place of x. The constants are bounded for $0 < \delta < 1/4$. The two marginal identities consequently bound the sum of the close-pair error integrals by $C\delta E$.

For a far pair, $0 \le D \le 1$ and $|\Phi| \le (|X(x)| + |X(y)|)/|x - y|$ bound the same two errors by

$$C \sum_{\substack{z \in \{x,y\} \\ d(z) > 0}} \left(d(z)^{L-1} + \frac{d(z)^L}{\ell(z)}\right).$$

The terms associated with x have $|x - y| > \ell(x)$. Applying the truncated-potential bound of 3.2 bounds their integrals by

$$C\int_{B_2} f(x)\left[\delta^{-\lambda} d(x)^{L-1-a_0\lambda} + \delta^{-\lambda-1} d(x)^{L-a_0(\lambda+1)}\right] dx.$$

With the integers fixed in 3.1, these powers are

$$L - 1 - a_0\lambda = a_0\sigma + 2N + 1 > 0, \qquad L - a_0(\lambda + 1) = a_0\sigma + N - 1 > 0.$$

The second positivity includes $N = 1$, since $\sigma > 0$. The source mass on B_2 is universally bounded by 3.2, and $0 \le d \le 1$. The displayed expression is thus at most a universal C_δ. The terms associated with y are identical with g instead of f.

Finally, the marginal identities give

$$E_f - E_g = \iint [\eta(x) - \eta(y)]\, d\pi,$$

and the subtraction is legitimate because the absolute integral is at most $E_f + E_g < \infty$. The close- and far-pair estimates show that the required sum of localization errors is at most $C\delta E + C_\delta$. Choosing δ sufficiently small proves the assertion for each prescribed $\varepsilon > 0$. □

5. The interval inequality

Lemma 5.1 (Interval comparison for Riesz kernels)**.** *Let $c > 0$, $\lambda > 0$, and $r > 0$. Set*

$$\mathcal{K}_r(s,s') = c(r^2 + (s-s')^2)^{-\lambda/2}, \qquad \mathcal{D}_r(s,s') = \frac{(s-s')^2}{r^2 + (s-s')^2}.$$

Alternatively, allow $r = 0$ when $0 < \lambda < 1$, with

$$\mathcal{K}_0(s,s') = c|s-s'|^{-\lambda}, \qquad \mathcal{D}_0(s,s') = 1$$

off the null diagonal; values on that diagonal do not affect the integrals below. Let $I, J \subset \mathbb{R}$ be nonempty open intervals, possibly unbounded, and let $c_I, c_J \ge 0$. Assume I is bounded whenever $c_I > 0$ and J is bounded whenever $c_J > 0$. For a bounded $I = (s_-, s_+)$ define

$$\mathcal{K}_{r,I}(s') = \frac{\mathcal{K}_r(s_-,s') + \mathcal{K}_r(s_+,s')}{2},$$

and define $\mathcal{K}_{r,J}(s)$ by the analogous average over the endpoints of J. Then

$$\int_{I\times J} (c_I + c_J)\mathcal{K}_r\, ds\, ds' \le \int_{I\times J} \big[c_I\mathcal{K}_{r,I}(s') + c_J\mathcal{K}_{r,J}(s) + \{\lambda \min(c_I,c_J)\mathcal{D}_r(s,s') + |c_I - c_J|\}\mathcal{K}_r(s,s')\big]\, ds\, ds'.$$

This is an inequality of nonnegative extended integrals. A term with zero endpoint coefficient is defined to be zero before its endpoints are evaluated.

Proof of 5.1. If at most one coefficient is positive, $|c_I - c_J| = c_I + c_J$, so the mismatch term alone equals the left integrand. This proves the assertion without subtracting possibly infinite quantities. Assume now both coefficients are positive. The two intervals are bounded.

First take $r > 0$, for which all integrals in this calculation are finite. Let m_I, m_J be the interval midpoints, $d_0 = m_I - m_J$, and $h = s - s'$. Integration by parts with the affine functions $s - m_I$ and $s' - m_J$ gives

$$\int_{I\times J} (2\mathcal{K}_r - \mathcal{K}_{r,I} - \mathcal{K}_{r,J}) = \lambda \int_{I\times J} \frac{h^2 - d_0 h}{r^2 + h^2}\mathcal{K}_r.$$

Indeed, for each fixed s',

$$\int_I (s - m_I)\partial_s\mathcal{K}_r(s,s')\, ds = |I|\mathcal{K}_{r,I}(s') - \int_I \mathcal{K}_r(s,s')\, ds.$$

The endpoint factor is $|I|/2$ at each end, which gives exactly the stated endpoint average. The analogous formula holds on J. Integrating in the other variable, adding, and using

$$\partial_s\mathcal{K}_r = -\frac{\lambda h}{r^2 + h^2}\mathcal{K}_r, \qquad \partial_{s'}\mathcal{K}_r = \frac{\lambda h}{r^2 + h^2}\mathcal{K}_r$$

gives the identity, since $(s - m_I) - (s' - m_J) = h - d_0$.

The pushforward of $ds\, ds'$ under $h = s - s'$ has density

$$\rho(h) = F(|h - d_0|), \qquad F(t) = \max\left\{0, \min\left\{|I|, |J|, \frac{|I| + |J|}{2} - t\right\}\right\}.$$

This is the overlap length $|I\cap(J+h)|$; the formula follows by intersecting two intervals centered a distance $|h-d_0|$ apart. In particular, it includes the case where one interval contains the other. The function F is nonincreasing. For $h>0$ and $d_0\ge 0$, $|h-d_0|\le|h+d_0|$, so $\rho(h)\ge\rho(-h)$; for $d_0\le 0$ the inequality reverses. Pairing positive and negative h therefore gives the absolutely convergent identity

$$\begin{aligned} d_0\int_{I\times J}\frac{h}{r^2+h^2}\mathcal{K}_r &= cd_0\int_0^\infty \frac{h[\rho(h)-\rho(-h)]}{(r^2+h^2)^{1+\lambda/2}}\,dh \\ &\ge 0. \end{aligned}$$

Consequently the sum of the two endpoint defects is at most $\lambda\int_{I\times J}\mathcal{D}_r\mathcal{K}_r$. Set $c_*=\min(c_I,c_J)$ and multiply that bound by c_*. Each individual defect $\int(\mathcal{K}_r-\mathcal{K}_{r,I})$ or $\int(\mathcal{K}_r-\mathcal{K}_{r,J})$ is at most $\int\mathcal{K}_r$, since the endpoint kernel is nonnegative. Multiply these individual bounds by c_I-c_* and c_J-c_*. Adding them to the common-coefficient bound and using

$$(c_I-c_*)+(c_J-c_*)=|c_I-c_J|$$

proves the claimed inequality at $r>0$.

For $r=0$, assume $0<\lambda<1$ and pass to the limit $r\downarrow 0$ in the final nonnegative inequality just proved. Off the diagonal, $\mathcal{K}_r\to\mathcal{K}_0$ and $\mathcal{D}_r\mathcal{K}_r\to\mathcal{K}_0$, with both functions bounded by $\mathcal{K}_0$. For every bounded interval H and every $a\in\mathbb{R}$,

$$\int_H|a-t|^{-\lambda}\,dt\le\frac{2}{1-\lambda}|H|^{1-\lambda}.$$

To see this, split H into the portions on either side of a. On each side the distances form an interval of length at most $|H|$, and integrating the decreasing function $t^{-\lambda}$ from zero gives the bound. Applying this estimate with $H=J$ and integrating over $a\in I$ proves integrability of $\mathcal{K}_0$ on the rectangle. Taking $H=J$ with $a\in\partial I$, or $H=I$ with $a\in\partial J$, proves integrability of every endpoint majorant, including coincident endpoints. Dominated convergence thus applies to all terms in the final inequality and gives the stated conclusion with $\mathcal{D}_0=1$ almost everywhere. □

6. One-dimensional pressure

Lemma 6.1 (Localized pressure in one dimension)**.** *In the setting of 3.1 with $N=1$, for every $\varepsilon>0$ there is a universal C_ε such that*

$$aW_f+bW_g\le\lambda\iint w(x)D(x,y)\,d\pi(x,y)+\varepsilon E+C_\varepsilon.$$

Proof of 6.1. All notation is from 3.1. Here $0<\sigma<1$, $\lambda=1-\sigma\in(0,1)$, and $D(x,y)=1$ off the null diagonal. The cutoff parameters are $a_0=4$, $k=5$, $L=8$.

The cutoff bounds proved in 4.2, specialized to $N=1$, give $w\in C^1(\mathbb{R})$ and

$$d^L\le\eta\le Cd^L,\quad 0\le w\le Cd^{L-1},\quad |w'|\le Cd^{L-2},\quad dw\le C\eta. \tag{O1}$$

Fix $0<\delta<1/4$, and put $\ell(x)=\delta d(x)^4$ when $d(x)>0$. Use the close and far pairs from the proof of 4.2, now with $a_0=4$. On a close pair, let z be an endpoint for which $|x-y|\le\ell(z)$. The segment comparability proved there, together with (O1), gives

$$|w(x)-w(y)|\le C\delta d(z)^{L-2+4}\le C\delta[\eta(x)+\eta(y)].$$

The marginal identities of 3.2 bound the close-pair integral by $C\delta E$. On far pairs, bound the difference by $w(x)+w(y)$. For every endpoint x in the support of w, the distance condition and the truncated-potential estimate of the same lemma give

$$\begin{aligned}\iint_{\text{far}} w(x)\,d\pi &\le C\int_{B_2}f(x)d(x)^{L-1}\ell(x)^{-\lambda}\,dx \\ &\le C_\delta\int_{B_2}f(x)d(x)^{L-1-4\lambda}\,dx\le C_\delta.\end{aligned}$$

The last exponent is $3+4\sigma>0$, and the source mass is universally bounded. The term with $w(y)$ is identical using g. Consequently

$$\iint |w(x)-w(y)|\,d\pi \le C\delta E + C_\delta. \tag{O2}$$

For $t>0$, every open source layer $\{f>t\}$ is a disjoint union of at most countably many open intervals. It is open even at a possible source singularity: the origin was assigned source value zero and thus is excluded, and every point of the layer has a neighborhood away from the origin on which f is continuous. Give each component I the constant weight

$$\chi_I = \begin{cases} \inf_I w, & I \text{ bounded and } |I| \le 2\delta \inf_I d, \\ 0, & \text{otherwise.} \end{cases}$$

Let $\chi_f(x,t)=\chi_I$ if $x\in I$, and set it to zero when $f(x)\le t$. Define $\chi_g(y,t')$ in the same way for the g layers. Thus $0\le\chi_f(x,t)\le w(x)$ and $0\le\chi_g(y,t')\le w(y)$.

These weights and the finite endpoints are measurable in the point and level. To verify this, let $T_+(x,t)$ and $T_-(x,t)$ be the forward and backward exit distances of the component through x, assigning both zero off the layer. For $r\ge 0$, the condition $T_+(x,t)>r$ is equivalent to $[x,x+r]\subset\{f>t\}$. Such a compact segment avoids zero, and $f-t$ has a strictly positive minimum on it. Thus the inclusion persists under small changes of x,t, proving measurability of T_+. The backward exit is treated identically. Their sum is the measurable component length, possibly infinite, and finite endpoints are $x_-=x-T_-$ and $x_+=x+T_+$. On the layer, for $\varphi=d$ or w, continuity gives

$$\inf_I \varphi = \inf_{s\in\mathbb{Q}} \begin{cases} \varphi(x+s), & -T_-(x,t)<s<T_+(x,t), \\ +\infty, & \text{otherwise.} \end{cases}$$

This countable infimum is measurable. The retention condition and χ_f are therefore measurable, and the same reasoning applies to χ_g.

Define the finite nonnegative losses

$$\mathcal{L}_f = \int_{\mathbb{R}} u(x) \int_0^{f(x)} [w(x)-\chi_f(x,t)]\,dt\,dx,$$

and $\mathcal{L}_g$ with v,g. Finiteness follows from $\mathcal{L}_f\le W_f$ and $\mathcal{L}_g\le W_g$, using 3.2. We next prove

$$\mathcal{L}_f+\mathcal{L}_g \le C\delta E + C_\delta. \tag{O3}$$

For x with $w(x)>0$, write $d=d(x)$ and $\ell=\delta d^4$. If the component I of an f layer through x has length at least ℓ, at least one side of I from x contains an interval of length $\ell/2$. That interval lies in B_3, since $|x|<2$ and $\ell<1/4$. Throughout it $f>t$. If C_0 is a universal upper bound on both $\int_{B_3} f$ and $\int_{B_3} g$, supplied by 3.2, then

$$t\,\ell/2 \le C_0, \qquad t\le (2C_0/\delta)d^{-4}.$$

Fix $P=1+2C_0/\delta$. Since $0<d\le 1$, at levels $t>Pd^{-5}$ this bound is impossible. Every component at such a level is therefore short: $|I|<\ell$. The Lipschitz bound gives $\inf_I d\ge(1-\delta)d$, and

$$|I|<\delta d^4\le\delta d\le 2\delta(1-\delta)d\le 2\delta\inf_I d.$$

Thus the retention condition holds. Regardless of whether $\inf_I w$ is positive or zero, (O1) along the component gives

$$w(x)-\chi_f(x,t)\le |I|\sup_I|w'|\le C\delta d^{L-2+4}\le C\delta\eta(x).$$

Integrating these high-level losses against $u(x)\,dt\,dx$ gives at most $C\delta E_f$. At lower levels,

$$\int_0^{\min(f(x),Pd^{-5})} [w(x)-\chi_f(x,t)]\,dt \le Pd^{-5}w(x)\le CPd^{L-1-5}\le CP,$$

where $L - 1 - 5 = 2$. Its integral against u is at most CP by the universal bound on $\int_{B_2} u$. Points with $w = 0$ contribute zero. Repeating this proof for g, v establishes (O3), with P now a fixed function of δ and equation parameters.

If $\chi_I > 0$, the closure of I is a compact subset of the annulus $1 < |x| < 2$, since w is continuous and $\inf_I w > 0$. At each endpoint x_-, x_+, continuity and maximality of the layer imply $f(x_\pm) = t$. Thus

$$u(x_\pm) = t^{1/q}|x_\pm|^{-B/q}.$$

For $x \in I$ let $\tau(x,t) = (t/|x|^B)^{1/q}$. Then $0 < \tau(x,t) < u(x)$, and the retention condition yields $|x_\pm - x| \le 2\delta d(x)$. The interval and its endpoints lie in the same component of $1 < |x| < 2$. Hence

$$\left|\log \frac{u(x_\pm)}{\tau(x,t)}\right| = \frac{|B|}{q}\,|\log|x_\pm| - \log|x|| \le \frac{2|B|}{q}\delta d(x).$$

Using $|e^z - 1| \le e^{|z|}|z|$, with $\delta < 1/4$ and $d \le 1$, gives a constant independent of δ and the solution. If u_I is the average of the two endpoint values, it follows that

$$|u_I - \tau(x,t)| \le C\delta d(x)\tau(x,t) \le C\delta d(x)u(x). \tag{O4}$$

The same argument, valid for every $p, q > 0$, gives $|v_J - \tau'(y,t')| \le C\delta d(y)v(y)$ on each positive-weight g component, with $\tau' = (t'/|y|^A)^{1/p}$. Products with zero component weight are defined to be zero before any endpoint is evaluated.

For fixed t, t', apply 5.1 to each pair of components $I \subset \{f > t\}$ and $J \subset \{g > t'\}$, with $r = 0$, $\lambda = 1 - \sigma \in (0,1)$, and coefficients χ_I, χ_J. The hypotheses hold because positive coefficients have bounded intervals. The factor $\mathcal{D}_0$ equals one almost everywhere, and $\min(\chi_I, \chi_J) \le w(x)$ on $I \times J$.

Sum over components and integrate over $t, t' > 0$. This uses only nonnegative measurable functions and Tonelli's theorem. In particular, the first endpoint integrand is defined by

$$\chi_f(x,t)\frac{K(x_- - y) + K(x_+ - y)}{2}$$

when $\chi_f > 0$, and zero otherwise; the finite exit maps proved measurable above make it a nonnegative Borel function of (x, y, t, t') on the layers. The partner endpoint term is defined analogously. Countable additivity over interval components requires no measurable enumeration. For each fixed (x,t) with positive weight, endpoint coincidences restrict y to at most two points. Measurability of the endpoint maps and Tonelli therefore make these coincidence sets null in the full product of points and levels. Their assigned kernel values do not affect the integrals. The interval lemma also supplies local integrability of the endpoint kernels because $\lambda < 1$. Integration over all partner levels and points gives the exact endpoint potentials. For instance, with selected $x_\pm$,

$$\int_{\mathbb{R}}\int_0^{g(y)} \frac{K(x_- - y) + K(x_+ - y)}{2}\,dt'\,dy = \frac{u(x_-) + u(x_+)}{2} = u_I.$$

No partner line or transverse diagonal is excluded in this one-dimensional argument.

All expressions after this integration are finite before any subtraction. The two left terms

$$U_f = \int u(x)\int_0^{f(x)} \chi_f(x,t)\,dt\,dx, \qquad U_g = \int v(y)\int_0^{g(y)} \chi_g(y,t')\,dt'\,dy$$

are bounded by W_f, W_g. The corresponding endpoint terms are bounded by CW_f, CW_g using (O4) and $\tau \le u$, $\tau' \le v$. The pressure term is at most λW_f, and the mismatch term is at most $W_f + W_g$, since $|\chi_f - \chi_g| \le w(x) + w(y)$ and the potential marginals are $f u\,dx$ and $g v\,dy$.

The absolute values of the signed differences below are bounded by the sums of the retained and endpoint integrands, so they too are integrable. We may therefore subtract the integrated

endpoint terms. Put

$$S = \int \int_0^{f(x)} \chi_f(x,t)[u(x) - u_I]\, dt\, dx + \int \int_0^{g(y)} \chi_g(y,t')[v(y) - v_J]\, dt'\, dy.$$

The sharper mismatch estimate

$$|\chi_f(x,t) - \chi_g(y,t')| \le |w(x) - w(y)| + [w(x) - \chi_f(x,t)] + [w(y) - \chi_g(y,t')]$$

gives the loss terms exactly: integration of $w(x) - \chi_f(x,t)$ over the partner point and level uses $u = K * g$ and produces $\mathcal{L}_f$; the other deficit produces $\mathcal{L}_g$ using $v = K * f$. Thus

$$S \le \lambda \iint w(x)\, d\pi + \iint |w(x) - w(y)|\, d\pi + \mathcal{L}_f + \mathcal{L}_g. \tag{O5}$$

The exact weighted layer identities are

$$\int_0^{f(x)} [u(x) - (t/|x|^B)^{1/q}]\, dt = af(x)u(x),$$

$$\int_0^{g(y)} [v(y) - (t'/|y|^A)^{1/p}]\, dt' = bg(y)v(y).$$

They follow by directly integrating the power of t; the values at the origin can be omitted. To compare the first identity with S, write $\tau = (t/|x|^B)^{1/q}$ and use (O4):

$$\begin{aligned} w(x)[u(x) - \tau] - \chi_f(x,t)[u(x) - u_I] &= [w(x) - \chi_f(x,t)][u(x) - \tau] + \chi_f(x,t)[u_I - \tau] \\ &\le [w(x) - \chi_f(x,t)]u(x) + C\delta d(x)w(x)u(x). \end{aligned}$$

With the zero-weight endpoint convention this estimate holds on all layers; when the coefficient is zero, only its first term is used. Integrate in t, x, apply $dw \le C\eta$, and repeat for g. This proves

$$aW_f + bW_g \le S + \mathcal{L}_f + \mathcal{L}_g + C\delta E. \tag{O6}$$

No sign was assumed for $u - u_I$ or $v - v_J$; only $u - \tau$ and $v - \tau'$ are nonnegative on their layers.

Combining (O5) and (O6) introduces two copies of $\mathcal{L}_f + \mathcal{L}_g$. Applying (O2) and (O3) gives

$$aW_f + bW_g \le \lambda \iint w(x)\, d\pi + C\delta E + C_\delta.$$

The coefficient C is independent of δ; the chosen threshold $P = 1 + 2C_0/\delta$ enters only the additive constant C_δ. Choose $\delta \in (0, 1/4)$ so that $C\delta \le \varepsilon$, and set $C_\varepsilon = C_\delta$. Since $D = 1$ almost everywhere, this is the claimed estimate. □

7. Pressure in higher dimensions and subcritical scaling

Lemma 7.1 (Localized pressure estimate for weighted sources)**.** *In the setting of 3.1, with $N \ge 2$, for every $\varepsilon > 0$ there is a universal C_ε such that*

$$aW_f + bW_g \le \lambda \iint w(x)D(x,y)\, d\pi(x,y) + \varepsilon E + C_\varepsilon.$$

Proof of 7.1. All notation is from 3.1. We use the source and potential estimates of 3.2 and the off-origin regularity in 4.1. Let $dS(e)$ be surface measure on $\mathbb{S}^{N-1}$ and write

$$d\nu(e) = \frac{dS(e)}{|\mathbb{S}^{N-1}|}.$$

The directional estimates below use normalized surface integration, polar coordinates, and Tonelli's theorem. Fix $0 < \kappa < 1$ and $0 < \delta < 1/4$, and let $\ell(x) = \delta d(x)^{a_0}$ when $d(x) > 0$. The parameters will be selected in the order κ, then δ, then P.

The cutoff bounds proved in 4.2 give $w \in C^1$ and

$$d^L \le \eta \le Cd^L, \qquad w \le Cd^{L-1}, \qquad |\nabla w| \le Cd^{L-2}, \qquad dw \le C\eta.$$

Choose a nonnegative smooth ψ on $[0,\infty)$, equal to one near zero, supported in $[0,1]$, and bounded by one. For $0<t\le 1$ and $\theta\in\mathbb{S}^{N-1}$ put

$$Z(t)=\int_{\mathbb{S}^{N-1}}\psi(|e-\theta|/t)\,d\nu(e).$$

Rotational invariance makes this independent of θ. Chordal polar coordinates give, including when $N=2$,

$$Z(t)=\frac{|\mathbb{S}^{N-2}|}{|\mathbb{S}^{N-1}|}t^{N-1}\int_0^1\psi(s)s^{N-2}(1-t^2s^2/4)^{(N-3)/2}\,ds.$$

Since $3/4\le 1-t^2s^2/4\le 1$ for $0\le s,t\le 1$, the integral factor and its derivative are uniformly bounded, including for the negative exponent when $N=2$. The condition $\psi=1$ near zero gives a strictly positive lower bound for that factor. Consequently $c_1t^{N-1}\le Z(t)\le C_1t^{N-1}$ and $|Z'(t)|\le C_1t^{N-2}$. Define

$$h_e(x)=\frac{w(x)\psi(|e-\omega_x|/(\kappa d(x)))}{Z(\kappa d(x))}\quad(1<|x|<2),\qquad h_e(x)=0\quad\text{otherwise.}$$

Its normalization is exactly $\int_{\mathbb{S}^{N-1}}h_e(x)\,d\nu(e)=w(x)$. Differentiation of the explicit expression gives

$$0\le h_e(x)\le C_\kappa d(x)^{L-N},\qquad |\nabla h_e(x)|\le C_\kappa d(x)^{L-N-1}.$$

For the derivative bound, set $\rho_e(x)=|e-\omega_x|/(\kappa d(x))$ on the annulus. Where $\psi'(\rho_e)\neq 0$, one has $|e-\omega_x|\le\kappa d(x)$, and $|\nabla\omega_x|\le C$ because $|x|>1$. Therefore

$$|\nabla\rho_e(x)|\le\frac{|\nabla\omega_x|}{\kappa d(x)}+\frac{|e-\omega_x||\nabla d(x)|}{\kappa d(x)^2}\le\frac{C_\kappa}{d(x)}.$$

The normalizer bounds give

$$\frac{|Z'(\kappa d(x))|}{Z(\kappa d(x))}\le\frac{C}{\kappa d(x)},\qquad\frac{|\nabla[Z(\kappa d(x))]|}{Z(\kappa d(x))}\le\frac{C}{d(x)}.$$

Together with $w\le Cd^{L-1}$ and $|\nabla w|\le Cd^{L-2}$, differentiating the three factors in h_e gives the asserted power d^{L-N-1}. The profile is constant near its center, so its apparent norm singularity causes no differentiability problem. The function extends with its first derivative by zero at both boundary spheres: w vanishes quadratically at radius one, and $L-N>1$ at radius two. These extensions are uniform in e, so $(e,x)\mapsto h_e(x)$ is jointly continuous.

Use the close and far pairs from the proof of 4.2, with the same $\ell(z)=\delta d(z)^{a_0}$. For a close pair choose an endpoint z with $|x-y|\le\ell(z)$. The segment comparability proved there allows us to integrate the derivative bound along the segment, giving

$$|h_e(x)-h_e(y)|\le C_\kappa\delta d(z)^{L-N-1+a_0}\le C_\kappa\delta d(z)^L.$$

The two marginals from 3.2 bound the integral over close pairs by $C_\kappa\delta E$. On a far pair, every endpoint z inside B_2 satisfies $|x-y|>\ell(z)$. Bound the difference by $h_e(x)+h_e(y)$ and apply the truncated-potential estimate to each endpoint. The x contribution is at most

$$C_\kappa\int_{B_2}f(x)d(x)^{L-N}\ell(x)^{-\lambda}\,dx\le C_{\kappa,\delta},$$

because $L-N-a_0\lambda=a_0\sigma+N+2>0$ and the source mass is bounded. The y contribution is the same with g. We have proved, uniformly in e,

$$\iint|h_e(x)-h_e(y)|\,d\pi\le C_\kappa\delta E+C_{\kappa,\delta}.\tag{P1}$$

On the support of $h_e(x)$, $|e-\omega_x|\le\kappa d(x)$, and therefore $|D_e(x,y)-D(x,y)|\le 2\kappa d(x)$. Surface integration and $dw\le C\eta$ give

$$\int_{\mathbb{S}^{N-1}}\left[\iint h_e(x)D_e(x,y)\,d\pi\right]d\nu(e)\le\iint w(x)D(x,y)\,d\pi+C\kappa E.\tag{P2}$$

Here C is independent of κ,δ.

For $0 < t < f(x)$ let I be the component containing x of the open set $\{f > t\}$ on the line $x + \mathbb{R}e$. This set is open despite the possible source singularity at zero: $f(0) = 0 < t$ and f is continuous off zero. Give it the intervalwise constant weight

$$\chi_I = \begin{cases} \inf_I h_e, & I \text{ bounded and } |I| \le 2\delta \inf_I d, \\ 0, & \text{otherwise.} \end{cases}$$

Write $\chi_f(e,x,t) = \chi_I$ when x lies in the layer, and set it to zero when $f(x) \le t$. Define $\chi_g(e,y,t')$ in the same way from the g layers. Thus $0 \le \chi_f \le h_e(x)$ and $0 \le \chi_g \le h_e(y)$. The exit-distance construction in the proof of 6.1 gives joint measurability also in the direction e. Indeed, inclusion of the compact segment $\{x + se : 0 \le s \le r\}$ in $\{f > t\}$ is open in (e,x,t): the segment avoids zero and has a strictly positive minimum of $f - t$. The backward segment has the same property. The infima defining the weights can again be taken over rational offsets between the measurable exits, now using the joint continuity of $(e,x) \mapsto h_e(x)$. Thus the component weights and finite endpoints are jointly Borel measurable.

Define the nonnegative finite losses

$$\mathcal{L}_f(e) = \int u(x) \int_0^{f(x)} (h_e(x) - \chi_f(e,x,t))\, dt\, dx,$$

and analogously $\mathcal{L}_g(e)$ with v, g. We claim that, for every $P \ge 1$,

$$\int_{\mathbb{S}^{N-1}} (\mathcal{L}_f(e) + \mathcal{L}_g(e))\, d\nu(e) \le (C_\kappa \delta + C_{\kappa,\delta} P^{-1}) E + CP. \tag{P3}$$

To prove it for f, u, fix $1 < |x| < 2$ and write $d = d(x)$ and $\ell = \delta d^{a_0}$. If the component I has length at most ℓ, it satisfies the retention condition because $\inf_I d \ge (1-\delta)d$ and $|I| \le \delta d \le 2\delta \inf_I d$. The preceding derivative estimate bounds its loss of weight by $C_\kappa \delta d^{L-N-1+a_0} \le C_\kappa \delta d^L$. Integrating these short-component losses over directions, levels and $u(x)dx$ gives at most $C_\kappa \delta E_f$.

The long components satisfy a pointwise cone estimate. Denote the component through x by $I_f(e,x,t)$, assigning it the empty set when $f(x) \le t$. If its length exceeds ℓ, one of the two exit distances from x exceeds $\ell/2$, including when the component is unbounded. Along that radial segment the source exceeds t. Hence for every direction e,

$$\mathbf{1}_{\{|I_f(e,x,t)| > \ell\}} \le \frac{N}{t(\ell/2)^N} \int_0^{\ell/2} [f(x + re) + f(x - re)] r^{N-1}\, dr. \tag{P4a}$$

Multiplication by $h_e(x)$, normalized surface integration and polar coordinates centered at x give

$$\begin{aligned} &\int_{\mathbb{S}^{N-1}} h_e(x) \mathbf{1}_{\{|I_f(e,x,t)| > \ell\}}\, d\nu(e) \\ &\quad \le \frac{2^N N}{|\mathbb{S}^{N-1}|\, t\ell^N} \int_{B_{\ell/2}(x)} f(y)[h_{\omega_{y-x}}(x) + h_{-\omega_{y-x}}(x)]\, dy \\ &\quad \le \frac{2^{N+1} N \sup_e h_e(x)}{|\mathbb{S}^{N-1}|\, t\ell^N} \int_{B_{\ell/2}(x)} f(y)\, dy \le \frac{C \sup_e h_e(x)}{t\ell^N}. \end{aligned} \tag{P4}$$

Here $B_{\ell/2}(x) \subset B_4$, whose source mass is universally bounded, and the choice of ω_{y-x} at $y = x$ has no effect. The radial segments avoid zero, so their source integrals are finite by off-origin continuity. Nonnegativity justifies Tonelli in the surface and radial variables. Both antipodal weights in (P4) are retained before taking the supremum, so the estimate applies to the localized angular weight without any symmetry assumption on h_e.

For $t \ge Pd^{-k}$, the surface integral of the long-component loss is bounded by the left side of (P4). Substituting the bound on h_e and $\ell = \delta d^{a_0}$ yields

$$\begin{aligned} &\int_{\mathbb{S}^{N-1}} (h_e(x) - \chi_f(e,x,t)) \mathbf{1}_{\{|I_f(e,x,t)| > \ell\}}\, d\nu(e) \\ &\qquad \le C_{\kappa,\delta} P^{-1} d^{L-N+k-a_0 N} = C_{\kappa,\delta} P^{-1} d^L, \end{aligned}$$

since $k = N(a_0 + 1)$. Integration over these levels against $u(x)dx$ gives at most $C_{\kappa,\delta}P^{-1}E_f$, because the level interval has length at most $f(x)$ and $d^L \le \eta$. For the remaining levels the exact normalization gives

$$\int_0^{\min(f(x),Pd^{-k})} \int_{\mathbb{S}^{N-1}} (h_e(x) - \chi_f(e,x,t))\, d\nu(e)\, dt \le Pd^{-k}w(x) \le CPd^{L-1-k} \le CP,$$

where $L-1-k = N+1 > 0$. Its integral against u is bounded by CP through 3.2. Combining this low-level bound with the short- and long-component bounds on the high levels, and repeating the argument for g, v, proves (P3).

It remains to turn the interval comparison into pressure. If $\chi_I > 0$, then I is bounded and its closure lies in the annulus where $h_e > 0$. At both endpoints x_-, x_+ continuity off zero and maximality of the layer component imply $f(x_-) = f(x_+) = t$. Consequently

$$u(x_\pm) = t^{1/q}|x_\pm|^{-B/q}.$$

For $x \in I$ set $\tau(x,t) = (t/|x|^B)^{1/q}$; the layer inequality gives $0 < \tau(x,t) < u(x)$. The retention condition gives $|x_\pm - x| \le 2\delta d(x)$. The segments from x to its endpoints lie in the closure of I, hence in the annulus. There $|\nabla \log |x|^{-B/q}| \le |B|/q$. The logarithmic comparison used for (O4) in the proof of 6.1 therefore applies along these segments. If u_I denotes the endpoint average, it gives

$$|u_I - \tau(x,t)| \le C\delta d(x)\tau(x,t) \le C\delta d(x)u(x). \tag{P7}$$

For a positive-weight g component J at level t', the endpoint average v_J satisfies the same estimate with $\tau'(y,t') = (t'/|y|^A)^{1/p}$ and $v(y)$ in place of τ, u. The constants may depend on A, B, p, q, which are fixed equation parameters. For zero weights all endpoint products are assigned value zero without evaluating an endpoint.

Fix e, t, t'. Write points on two distinct parallel lines as $x = z + se$, $y = z' + s'e$, with $z, z' \in e^\perp$ and $z \ne z'$. For fixed e this orthogonal decomposition has Jacobian one. Restriction of K gives the kernel of 5.1 with $r = |z - z'|$ and exponent λ. Apply that lemma to every pair of components of the two layers, with coefficients χ_I, χ_J. Bound $\min(\chi_I, \chi_J) \le h_e(x)$.

The resulting endpoint integrands can be defined directly from the measurable finite exit maps $(e,x,t) \mapsto x_-, x_+$: when $\chi_f > 0$, use $\chi_f[K(x_- - y) + K(x_+ - y)]/2$, and otherwise set that term to zero before evaluating either endpoint. Define the partner endpoint term in the same manner. These are nonnegative Borel functions on the layer variables. Each open line section is the disjoint union of at most countably many intervals, so summation over its components is the countable additive decomposition of the corresponding integral. No measurable enumeration of components is needed. Tonelli therefore applies directly to the nonnegative integrands in (x, y, t, t'). All final integrands use the original variables e, x, y and the exit maps; later surface integration requires no globally continuous choice of coordinates in $e^\perp$.

The excluded same-line set can be written without transverse coordinates as

$$\{(e,x,y) : (\mathrm{Id} - e \otimes e)(x - y) = 0\}.$$

It is Borel. For each fixed e, x, its y-section is an affine line and has zero N-dimensional measure because $N \ge 2$. Tonelli makes it null also in the joint direction and point variables. The two left terms become

$$U_f(e) = \int \int_0^{f(x)} \chi_f(e,x,t)u(x)\, dt\, dx, \quad U_g(e) = \int \int_0^{g(y)} \chi_g(e,y,t')v(y)\, dt'\, dy.$$

The endpoint terms become the same expressions with u_I or v_J in place of $u(x)$ or $v(y)$. Indeed, integrating the endpoint kernel against all partner points and levels recovers the endpoint potential by 3.2. For each selected endpoint the excluded partner points on its affine line have zero N-dimensional measure and do not change that potential.

We verify finiteness before subtracting these expressions. Let $H_f(e) = \int h_e f u$ and $H_g(e) = \int h_e g v$, which are finite because h_e is bounded and compactly supported. Then $U_f \le H_f$ and $U_g \le H_g$. The endpoint terms are at most CH_f, CH_g by (P7) and $\tau \le u$, $\tau' \le v$ on their respective layers. The pressure term is at most λH_f. The coefficient mismatch is bounded by

$h_e(x)+h_e(y)$, so its integrated term is at most H_f+H_g by the potential marginals. All resulting integrals are finite even if individual unweighted integrals along unbounded lines diverge.

After subtraction, use the sharper bound

$$|\chi_f(e,x,t)-\chi_g(e,y,t')|\le |h_e(x)-h_e(y)|+(h_e(x)-\chi_f)+(h_e(y)-\chi_g).$$

The integrated comparison therefore yields

$$S_e\le \lambda\iint h_e(x)D_e(x,y)\,d\pi+\iint |h_e(x)-h_e(y)|\,d\pi+\mathcal{L}_f(e)+\mathcal{L}_g(e), \tag{P5}$$

where

$$S_e=\int\int_0^{f(x)}\chi_f(e,x,t)(u(x)-u_I)\,dt\,dx+\int\int_0^{g(y)}\chi_g(e,y,t')(v(y)-v_J)\,dt'\,dy.$$

The weighted layer identities and the comparison algebra leading to (O6) in the proof of 6.1 are pointwise in the space variable and do not depend on its dimension. Apply them with w replaced by h_e, using $0\le\chi_f\le h_e$, $\tau\le u$, and (P7), and likewise for g,v. Retaining the endpoint errors before angular integration gives

$$aH_f(e)+bH_g(e)\le S_e+\mathcal{L}_f(e)+\mathcal{L}_g(e)+C\delta\int d(x)h_e(x)f(x)u(x)\,dx+C\delta\int d(y)h_e(y)g(y)v(y)\,dy. \tag{P6}$$

No sign is assumed for the differences $u-u_I$ or $v-v_J$; only the ideal layer differences $u-\tau$ and $v-\tau'$ are nonnegative.

The signed term is absolutely integrable in the direction variable: the retained and endpoint bounds above and angular normalization give

$$\int_{\mathbb{S}^{N-1}}|S_e|\,d\nu(e)\le C\int_{\mathbb{S}^{N-1}}(H_f(e)+H_g(e))\,d\nu(e)=C(W_f+W_g)<\infty.$$

Thus we may combine (P5) and (P6) and integrate against $d\nu(e)$. There are two copies of the layer losses. Equations (P1), (P2), and (P3), the normalization $\int_{\mathbb{S}^{N-1}}h_e\,d\nu=w$, and the bound $dw\le C\eta$ for the two endpoint errors give

$$aW_f+bW_g\le\lambda\iint wD\,d\pi+(C\kappa+C\delta+C_\kappa\delta+C_{\kappa,\delta}P^{-1})E+C_{\kappa,\delta,P}.$$

First choose κ small, then δ small with κ fixed, and finally P large. The coefficient of E becomes at most the prescribed ε, proving the lemma. □

Theorem 7.2 (Weighted subcritical nonexistence for Riesz systems)**.** *Let $N\ge1$ be an integer, let $0<\sigma<N$, let $c>0$, let $A,B>-\sigma$ be real weights, and let $p,q>0$ satisfy $pq>1$ and*

$$\frac{N+A}{p+1}+\frac{N+B}{q+1}>N-\sigma.$$

There are no finite continuous functions $u,v:\mathbb{R}^N\to(0,\infty)$ satisfying, at every $x\in\mathbb{R}^N$,

$$u(x)=c\int_{\mathbb{R}^N}|x-y|^{\sigma-N}|y|^Av(y)^p\,dy,\qquad v(x)=c\int_{\mathbb{R}^N}|x-y|^{\sigma-N}|y|^Bu(y)^q\,dy.$$

Proof of 7.2. Suppose a solution exists and use 3.1. Set

$$\gamma=a(N+B)+b(N+A)-\lambda>0,$$

$$T=\iint w(x)D(x,y)\,d\pi,\qquad Q=\iint(\Phi(x,y)-\eta(x)+w(x)D(x,y))\,d\pi.$$

All these integrals are finite by 3.2 and 4.2. The virial identity and the marginal $\int\eta(x)d\pi=E_f$ give exactly

$$c_1E_f+c_2E_g=aW_f+bW_g-\lambda T+\lambda Q,\qquad c_1=a(N+B)-\lambda,\quad c_2=b(N+A).$$

For arbitrary $\varepsilon > 0$, 7.1 (if $N \geq 2$) or 6.1 (if $N = 1$) bounds the first three terms on the right by $\varepsilon E + C_\varepsilon$, while 4.2 gives $|Q| + |E_f - E_g| \leq \varepsilon E + C_\varepsilon$, after taking a common universal constant. No sign condition on c_1 or c_2 is needed: since $c_1 + c_2 = \gamma$,

$$c_1 E_f + c_2 E_g = \frac{\gamma}{2} E + \frac{c_1 - c_2}{2}(E_f - E_g).$$

Consequently, with $M = 1 + \lambda + |c_1 - c_2|/2 > 0$,

$$\frac{\gamma}{2} E \leq M\varepsilon E + M C_\varepsilon.$$

Choose $\varepsilon_* = \gamma/(4M) > 0$ and subtract the resulting $\gamma E/4$ from both sides. Since E is finite for each solution, this gives the universal upper bound $E \leq 4MC_{\varepsilon_*}/\gamma$. Because $\eta = 1$ on B_1, it follows that

$$\int_{B_1} (|x|^B u^{q+1} + |x|^A v^{p+1}) \leq C.$$

For each $R > 0$ define $u_R(x) = R^\alpha u(Rx)$ and $v_R(x) = R^\beta v(Rx)$. The dilation calculation in the proof of 3.2 shows that they are finite continuous positive solutions of the same system with the same parameters, including c. Apply the universal energy bound to this pair. Both weighted energies have the same scaling exponent, since

$$\alpha(q+1) - B = \beta(p+1) - A = \alpha + \beta + \sigma.$$

Thus

$$\int_{B_1} (|x|^B u_R^{q+1} + |x|^A v_R^{p+1}) = R^{\alpha+\beta+\sigma-N} \int_{B_R} (|z|^B u(z)^{q+1} + |z|^A v(z)^{p+1})\, dz.$$

The algebraic identity

$$\gamma = (1 - a - b)(\alpha + \beta + \sigma - N), \qquad 1 - a - b = \frac{pq - 1}{(p+1)(q+1)} > 0$$

shows that the exponent is strictly positive. For $R \geq 1$ the last integral is at least its fixed positive value on B_1. The right side therefore tends to infinity as $R \to \infty$, contradicting the universal bound. This proves the theorem. □

8. Completion of the main theorem

Proof of 1.1. The hypotheses place the pair in 2.1. If $pq \leq 1$, 2.4 already gives the zero pair, without using the strict gap. Suppose $pq > 1$. By 2.3, either the pair is zero or u, v are strictly positive finite continuous functions, $A, B > -2s$, and the normalized exact Riesz system holds at every point. In the latter alternative apply 7.2 with $\sigma = 2s$ and $c = c_{N,s} > 0$. All its hypotheses hold: $N \geq 1$, $0 < 2s < N$, the derived weight bounds, positive exponents of product greater than one, and precisely the assumed weighted gap. This excludes the positive alternative and proves $u \equiv v \equiv 0$. □

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